



Modelling Financial Markets

Chapter 19: Review and Project Presentations



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CV: Cross-Validation
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ EVT: Extreme Value Theory
- ▣ FDR: False Discovery Rate
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GBM: Geometric Brownian Motion
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HS: Historical Simulation
- ▣ IGARCH: Integrated GARCH
- ▣ IQR: InterQuartile Range
- ▣ LRMEs: Long-Run Marginal Expected Shortfall
- ▣ MCS: Model Confidence Set
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SML: Security Market Line
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ VOLARE: VOLatility Archive for Realized Estimates (database)

Outline

Introduction

Review I: Markets, Returns and Pricing

Review II: Volatility, Dependence and Risk

Review III: High Frequency, Continuous Time and Options

Review IV: Learning, Text and New Markets

Common Pitfalls

Integrated Case Study: BET

The Team Project

Using AI in Your Project

Project Seeds

Case Study: Nonstandard Errors

AI for Scientific Discovery

Conclusions

References

What This Chapter Does

Purpose

- Review Chapters 0–18 as one connected story
- Collect the pitfalls that returned in every chapter
- Run one integrated case study, from raw prices to a backtested value-at-risk (VaR) at 1%
- Prepare the team project: its four steps, the question, the methods and the AI policy

Assessment

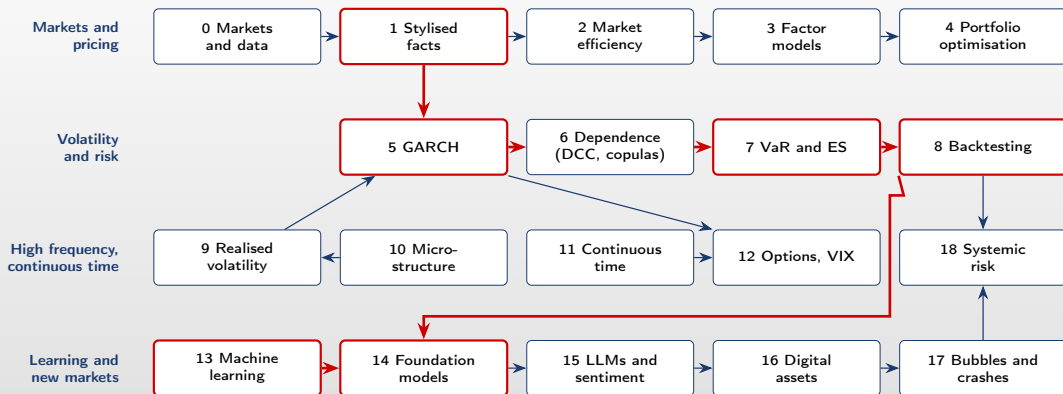
- Team project (GitHub, presentation, oral defence): **70%**
- Quizzes and seminar activity: **20%**
- Attendance: **10%**; there is no final exam



Bucharest University of Economic Studies (ASE), Piața Romană

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The Course in One Picture



- Red: the main thread of the course, from stylised facts to risk forecasts that we can test
- Blue arrows: where one chapter supplies the input or the benchmark of another

Reading the Course Map

- Chapter 1 sets the targets: heavy tails, volatility clustering, almost no linear predictability
 - ▶ Every later model is judged by whether it reproduces these facts
- Chapters 5–8 turn the facts into a risk number and a test of that number
 - ▶ GARCH gives σ_{t+1} ; VaR and ES give the loss; backtesting says whether to trust it
- Chapters 9–12 measure volatility better (intraday data, option prices) and in continuous time
- Chapters 13–15 bring flexible models, always evaluated with the tools of Chapter 8
- Chapters 16–18 apply the same toolkit to crypto, bubbles and the whole financial system
- Special Chapter 20: forecasting realised volatility with path signatures on the VOLARE data

Five Questions for Every Model

Before estimating

- **Data:** which series, which calendar, which adjustments (dividends, splits, holidays)?
- **Model:** which stylised fact does it capture, and which does it ignore?
- **Estimation:** on which window, with which information at time t ?

After estimating

- **Evaluation:** out of sample, with a loss that matches the target (QLIKE, FZ0) and the right diagnostic tests (e.g. Kupiec)?
- **Uncertainty:** robust standard errors, bootstrap, a test against a simple benchmark?
- If one answer is missing, the result is not yet a finding

Chapter 0: Financial Markets in 2026

Key idea

- Markets combine exchanges, over-the-counter dealers, exchange-traded funds (ETFs), algorithms and blockchains
- Data quality is a modelling decision: source, adjustment and calendar change the result

Key inference

- For i.i.d. log returns, annualised by the number of periods a year: $se(\hat{\mu}_{\text{ann}}) = \sigma_{\text{ann}} / \sqrt{Y}$ over Y years, whatever the frequency; $se(\hat{\sigma}) \approx \sigma \sqrt{(K-1)/(4T)}$ over T days, K the population kurtosis (finite fourth moment needed); serial dependence calls for long-run variances
- SPY 2015–2026: $se(\hat{\mu}_{\text{ann}}) \approx 5.1\%$ for a mean of 13.7%: means are barely estimable, volatilities are

Result from the data

- 2015–2026: SPY (S&P 500 ETF, dividends reinvested) 13.7% a year with 17.6% volatility; Bitcoin 60.6% a year with 66.7%
- BET, July 2007 to February 2009: a drawdown of −82.5%

Leads to

- Returns, not prices, are the object of study (Chapter 1)

Chapter 1: Stylised Facts and Returns

Key idea

- ▣ Returns are uncorrelated but not independent: their magnitude is predictable, but their sign is largely unpredictable
- ▣ Heavy tails and volatility clustering in every market [Cont \(2001\)](#)

Key inference

- ▣ Ljung–Box $Q(m) \sim \chi_m^2$ needs i.i.d. returns; under volatility clustering it over-rejects: use the robust Q^* with $\hat{\rho}_k^2$ weighted by $T\hat{\rho}_k^2/\hat{\tau}_k$ [Lobato, Nankervis & Savin \(2001\)](#)
- ▣ Tails: Hill $\hat{\alpha}$ with $se \approx \hat{\alpha}/\sqrt{k}$ [Hill \(1975\)](#); if $\alpha < 4$ the sample kurtosis is not consistent and its bootstrap fails [Athreya \(1987\)](#)

Result from the data

- ▣ BET, Hill index of losses (top $k = 153$): $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03]: the fourth moment is likely infinite
- ▣ Days beyond 5σ are more than 1000 times more frequent than the Normal distribution predicts

Leads to

- ▣ Volatility models (Chapter 5) and tail risk measures (Chapter 7)

Chapter 2: Market Efficiency

Key idea

- Efficiency: prices fully reflect the information set; a test of abnormal-return predictability is a joint test of efficiency and the expected-return model [Fama \(1970\)](#)
- Efficiency varies over time (adaptive markets); tests need robust inference

Key inference

- Heteroskedasticity-robust variance ratio $z^*(q) \Rightarrow N(0, 1)$ under a martingale difference null [Lo & MacKinlay \(1988\)](#)
- Several horizons q at once: the joint test $\max_q |z^*(q)|$ with studentised maximum modulus critical values [Chow & Denning \(1993\)](#); rolling windows need a multiplicity correction too

Result from the data

- Share of rolling windows that reject the random walk: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- Statistical rejection is not exploitable profit

Leads to

- Anomalies and factors (Chapter 3); the multiple-testing problem

Chapter 3: Factor Models and Asset Pricing

Key idea

- Expected returns should reward only exposure to systematic factors
- Hundreds of published factors require higher hurdles Harvey, Liu & Zhu (2016)

Key inference

- $$GRS = \frac{T-N-K}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{\mathbf{1} + \hat{\mu}_f^\top \hat{\Omega}^{-1} \hat{\mu}_f} \sim F(N, T-N-K) \text{ only}$$
 under i.i.d. Normal errors Gibbons, Ross & Shanken (1989)
- Fama–MacBeth s.e. ignore that betas are estimated: with $c = \hat{\gamma}^\top \hat{\Sigma}_f^{-1} \hat{\gamma}$, Shanken's covariance is $(1+c)(\hat{V}_{FM} - \hat{\Sigma}_f/T) + \hat{\Sigma}_f/T$ for estimated factor loadings Shanken (1992)

Result from the data

- 25 size and book-to-market portfolios: SML slope -4.4% , $GRS = 4.18$: the CAPM is rejected
- Sector ETFs: the first principal component (the market) explains 66% of the variance

Leads to

- Portfolio construction (Chapter 4); factor models inside machine learning (Chapter 13)

Chapter 4: Portfolio Optimisation

Key idea

- Markowitz asks the right question, but estimation error dominates the answer [Markowitz \(1952\)](#)
- Shrinkage, constraints and risk-based rules stabilise the weights; $1/N$ is a hard benchmark [DeMiguel, Garlappi & Uppal \(2009\)](#)

Key inference

- $\text{se}(\widehat{SR}) \approx \sqrt{(1 + \widehat{SR}^2/2)/T}$ for i.i.d. Normal returns, per period [Lo \(2002\)](#); serial correlation changes the annualisation factor
- Is portfolio A better than $1/N$? Test $SR_A - SR_B = 0$ with HAC or studentised bootstrap inference [Ledoit & Wolf \(2008\)](#), not by comparing two point estimates

Result from the data

- 60-month samples: median in-sample Sharpe 1.57, median true Sharpe 0.30
- 96% of the estimated tangency portfolios are truly worse than $1/N$

Leads to

- Risk budgets need volatility and dependence forecasts (Chapters 5–6)

Recap: Markets, Returns and Pricing

- ▣ Clean data first: adjusted prices for ETFs and stocks, own calendar for each series, join on prices before returns
- ▣ Stylised facts are the benchmark for every model that follows
- ▣ Efficiency and factor tests need robust errors and multiple-testing corrections
- ▣ Optimisation amplifies estimation error: evaluate out of sample, against $1/N$

Chapter 5: Conditional Volatility: GARCH

Key idea

- Volatility clusters, so today's shock predicts tomorrow's variance [Engle \(1982\)](#), [Bollerslev \(1986\)](#)
- Persistence close to 1; asymmetry in equities; heavy-tailed innovations

Key inference

- QMLE: $\sqrt{T}(\hat{\theta} - \theta) \Rightarrow N(0, A^{-1}BA^{-1})$ (sandwich); Hessian-only s.e. are wrong for non-Normal innovations [Bollerslev & Wooldridge \(1992\)](#)
- Near $\alpha + \beta = 1$ the parameter is close to the boundary of the covariance-stationary region: Wald intervals are unreliable; use profile likelihood or bootstrap restricted to the parameter space [Andrews \(1999\)](#)

Result from the data

- S&P 500, 2000–2026: $\alpha + \beta = 0.9947$, a half-life of 131 days
- BET: profile-likelihood 95% CI for $\alpha + \beta$ is $[0.975, 1]$ (IGARCH not rejected): the half-life is at least 27 days, with no finite upper bound

Leads to

- $\hat{\sigma}_{t+1}$ is the input of filtered VaR and ES (Chapter 7)

Chapter 6: Multivariate Volatility and Dependence

Key idea

- ▣ Correlations move with the macro regime; diversification is not guaranteed
- ▣ Copulas separate the margins from dependence; tail dependence drives joint losses

Key inference

- ▣ DCC is estimated in two steps: second-step s.e. must account for the first-step volatilities [Engle \(2002\)](#)
- ▣ Crisis vs calm correlation: ρ rises mechanically with volatility; test on $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, δ the relative rise in variance [Forbes & Rigobon \(2002\)](#)

Result from the data

- ▣ SPY–TLT (long US Treasury bond ETF): ρ from -0.40 to 0.11 after 2022, so a 60/40 portfolio went from 10.4% to 13.2% volatility
- ▣ Bitcoin–SPY DCC correlation: 0.04 before 2020, 0.31 since

Leads to

- ▣ Portfolio VaR and Euler contributions (Chapter 7); systemic risk (Chapter 18)

Chapter 7: Value-at-Risk and Expected Shortfall

Key idea

- VaR is a quantile of the loss, ES the average loss beyond it; ES is coherent
- Filtered historical simulation and conditional EVT react to volatility in days McNeil & Frey (2000)

Key inference

- Historical VaR, i.i.d. returns with a continuous density $f(q_\alpha) > 0$: $\sqrt{T}(\hat{q}_\alpha - q_\alpha) \Rightarrow N(0, \alpha(1 - \alpha)/f(q_\alpha)^2)$; in the tail f is small, so the estimate is noisy; under dependence $\alpha(1 - \alpha)$ becomes the long-run variance of the hits $\mathbb{1}\{r_t \leq q_\alpha\}$
- ES averages the αT largest losses: with αT of a few dozen and a tail index near 3, its variance is dominated by one or two days

Result from the data

- S&P 500, daily: historical $\text{VaR}_{1\%} = 3.45\%$, $\text{ES}_{1\%} = 5.01\%$
- The Normal distribution gives only 2.80%: heavy tails matter

Leads to

- A risk forecast is useful only if it survives a backtest (Chapter 8)

Chapter 8: Backtesting and Evaluating Risk Forecasts

Key idea

- Test the number and the timing of breaches: coverage and independence [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- Rank (VaR, ES) forecasts with a consistent score (FZ0) and Diebold–Mariano tests [Fissler & Ziegel \(2016\)](#)

Key inference

- Kupiec $LR_{uc} \sim \chi_1^2$, Christoffersen $LR_{ind} \sim \chi_1^2$; the DQ test regresses $I_t - \alpha$ on lagged hits and VaR_t , $DQ \sim \chi_k^2$ [Engle & Manganelli \(2004\)](#)
- Power is low: at $T = 250$ the 5% Kupiec test detects a true rate of 2% only 24% of the time; compare models with conditional predictive ability tests [Giacomini & White \(2006\)](#)

Result from the data

- S&P 500, VaR 1%, 2007–2026: Normal 2.70%, HS 1.67%, FHS 1.23%, GARCH-EVT 1.13% breaches
- A GARCH filter plus a flexible tail wins in all four markets

Leads to

- Forecast-comparison tests (DM, MCS) with a suitable loss also judge realised-volatility forecasts; coverage tests judge quantile forecasts, including machine-learning ones (Chapters 9, 13, 14)

Recap: Volatility, Dependence and Risk

- ▣ GARCH turns volatility clustering into one-day-ahead forecasts
- ▣ Dependence is dynamic and strongest in the tails; crisis correlations need a volatility adjustment
- ▣ VaR and ES must combine a volatility filter with a heavy-tailed distribution
- ▣ A risk model is accepted only after tests of coverage, independence and ES

Chapter 9: Realised Volatility

Key idea

- Intraday prices measure daily variance (RV , realised variance); the log-HAR (heterogeneous autoregressive) model captures its long memory [Corsi \(2009\)](#)
- Returns divided by realised volatility have a distribution close to the Normal distribution

Key inference

- HAR errors are heteroskedastic and, for multi-day targets that overlap, serially correlated: use HAC s.e.; forecasts are compared with DM tests on a robust loss [Patton \(2011\)](#)
- Microstructure noise biases RV upwards at high frequency: two-scale estimators remove the bias [Zhang, Mykland & Aït-Sahalia \(2005\)](#)

Result from the data

- SPY: kurtosis 15.73 for open-to-close returns, 2.65 after dividing by $\sqrt{RV_t}$
- log-HAR beats GARCH(1,1)- t in QLIKE by 10% for SPY ($p = 0.021$) and 22% for Bitcoin

Leads to

- Better volatility inputs for VaR; microstructure explains the noise (Chapter 10)

Chapter 10: Market Microstructure

Key idea

- Prices form in limit order books; the spread pays for processing, inventory and adverse selection
- Liquidity follows the clock and the VIX, and disappears in stress

Key inference

- The Roll spread needs $\text{Cov}(\Delta p_t, \Delta p_{t-1}) < 0$; when the sample covariance is positive the estimator is undefined, so report how often
- An illiquidity premium is a cross-sectional claim: Fama–MacBeth with Newey–West s.e. and the Shanken correction [Shanken \(1992\)](#)

Result from the data

- Amihud ratio (bp per USD 1 million): SPY 0.0016, Banca Transilvania 24.7, Transelectrica 1024.0
- SPY: the opening 5-minute bar moves 2.06 times as much as a midday bar

Leads to

- Liquidity risk in VaR and in the project data choices

Chapter 11: Continuous-Time Models

Key idea

- Brownian motion is the limit of random walks; Itô's lemma solves the price equations
- Jumps explain the tails, stochastic volatility the clustering

Key inference

- Normality of GBM log returns: Jarque–Bera $\sim \chi^2_2$ rejects everywhere; jumps are tested with bipower variation, $(RV - BV)$ standardised $\Rightarrow N(0, 1)$ [Barndorff-Nielsen & Shephard \(2006\)](#)
- The test is daily: over many days, control the number of false jump days

Result from the data

- S&P 500 against geometric Brownian motion: 62 days beyond ± 4 standard deviations, against 0.59 expected
- Hill index of S&P 500 losses: 2.87 [2.07, 3.67] (Chapter 16): the fourth moment is likely infinite, unlike the Normal log returns of GBM

Leads to

- Option pricing needs only σ , not μ (Chapter 12)

Chapter 12: Options and the Volatility Surface

Key idea

- Black–Scholes prices options by hedging; implied volatility is the market's view of σ [Black & Scholes \(1973\)](#)
- Smile and skew encode the fear of crashes

Key inference

- VRP_t uses the next 21 days of realised variance: consecutive values overlap, so the mean needs Newey–West s.e. with at least 21 lags
- Implied volatility is a risk-neutral forecast: the premium, not a forecast error, explains most of the gap

Result from the data

- 1990–2026: mean VIX 19.4, mean realised volatility over the next month 15.3
- The variance risk premium is positive on 86% of days, but not in crashes

Leads to

- Implied volatility as a feature for machine learning and as a stress indicator (Chapters 13, 18)



Recap: High Frequency, Continuous Time and Options

- ▣ Intraday data measure volatility; log-HAR forecasts it better than daily GARCH
- ▣ Liquidity is a risk of its own and varies by orders of magnitude across markets
- ▣ Continuous-time models are tractable, but realistic ones need jumps and random volatility
- ▣ Option prices carry forward-looking information, including a variance risk premium

Chapter 13: Machine Learning in Finance

Key idea

- Flexible models, but little signal and much noise; the finance-specific part is the process [López de Prado \(2018\)](#)
- Purged cross-validation, meta-labelling and deflated Sharpe ratios [Bailey & López de Prado \(2014\)](#)

Key inference

- Selection among N strategies: the deflated Sharpe ratio tests $SR > E[\max_N \widehat{SR}]$ under the null of no skill [Bailey & López de Prado \(2014\)](#)
- Cross-validation on time series must purge overlapping labels and embargo the test fold, or its accuracy is biased upwards

Result from the data

- Shuffled K-fold reports 69.4% accuracy on pure noise: leakage
- S&P 500: the fractionally differenced series is stationary at $d^* = 0.25$; no model beats “always up”

Leads to

- Deep learning and foundation models, judged the same way (Chapter 14)

Chapter 14: Deep Learning and Time-Series Foundation Models

Key idea

- LSTMs (long short-term memory networks) and Transformers learn from sequences; foundation models forecast zero-shot
- Evaluate them with Chapter 8 tools, after the release of their weights

Key inference

- DM test on QLIKE differences with HAC variance: $t_{DM} \Rightarrow N(0, 1)$ for non-nested comparisons [Diebold & Mariano \(1995\)](#)
- With many candidate models, the model confidence set controls the family-wise error [Hansen, Lunde & Nason \(2011\)](#)

Result from the data

- SPY realised volatility, QLIKE: Chronos-2 0.231, log-HAR 0.234, GARCH- t 0.283; the first two are not significantly different
- Raw Chronos-2 VaR 1% on the S&P 500: 1.67% breaches

Leads to

- Text as data (Chapter 15)

Chapter 15: LLMs and Sentiment Analysis

Key idea

- From dictionaries to FinBERT and large language models (LLMs): text becomes a number
- Text embeddings enter forecasting models as features

Key inference

- Predictive regression $r_{t+1} = a + bS_t + \varepsilon_{t+1}$: a persistent S_t whose innovations correlate with returns biases $\hat{b}$ and the t -test [Stambaugh \(1999\)](#)
- Classifier accuracies on the same texts are paired: compare them with the exact McNemar test, not two separate intervals

Result from the data

- Qwen 7B vs 1.5B on financial headlines: +21.5 percentage points in accuracy, McNemar $p < 0.001$
- A headline-sentiment strategy: Sharpe +0.63 before and -1.48 after costs; break-even cost 1.5 bp per trade

Leads to

- Sentiment in crypto markets and bubbles (Chapters 16–17)

Chapter 16: Digital Assets and DeFi

Key idea

- ▣ Cryptocurrencies form a distinct asset class in the feature space of [Pele et al. \(2023\)](#), where a tail factor separates them; their Hill tail indices are not significantly lower than those of equities or gold (Chapter 16)
- ▣ Stablecoins, automated market makers and spot ETFs connect crypto to mainstream finance

Key inference

- ▣ Crypto factor models: GRS and Fama–MacBeth with the Shanken correction [Gibbons, Ross & Shanken \(1989\)](#), [Shanken \(1992\)](#); heavy tails ask for HAC or bootstrap versions
- ▣ Tails: Hill indices with intervals, not sample kurtosis (Chapter 1); a break in a correlation is tested with an unknown date [Andrews \(1993\)](#)

Result from the data

- ▣ Bitcoin–S&P 500 link: sup-Wald 20.2 (5% critical value 11.6), break estimated on 9 Mar 2020, not at the 2024 ETF launch
- ▣ Hill index of losses: Bitcoin 2.73 [2.12, 3.33], S&P 500 2.87 [2.07, 3.67]: statistically indistinguishable

Leads to

- ▣ Explosive price episodes (Chapter 17)

Chapter 17: Bubbles and Crashes

Key idea

- ▣ A bubble is a price path that grows faster than fundamentals can justify
- ▣ Right-tailed unit-root tests date the start and the end of explosive episodes

Key inference

- ▣ Under the unit-root null, SADF and GSADF converge to functionals of Brownian motion; critical values depend only on the minimum window r_0 Phillips, Shi & Yu (2015)
- ▣ Volatility shifts distort the size: use wild-bootstrap critical values; control the multiplicity of date-stamping

Result from the data

- ▣ Bitcoin 2014–2026 (weekly): GSADF 3.85 against a 95% critical value of 2.27: explosive episodes are detected
- ▣ One vs two Markov regimes for Bitcoin: $LR = 123.2$ against a parametric-bootstrap 95% critical value of 11.8 ($p = 0.005$)

Leads to

- ▣ Contagion from one market to the system (Chapter 18)

Chapter 18: Systemic Risk and Networks

Key idea

- Systemic risk: losses of one institution that spread to the whole system
- CoVaR (the VaR of the system conditional on the distress of one institution), SRISK and spillover networks measure the contribution of each institution

Key inference

- ΔCoVaR differences between banks: paired tests on joint block-bootstrap draws with a Holm correction; at $\alpha = 1\%$ only about 40 tail days, so extremal-quantile issues
- SRISK: LRMES simulated from GJR-GARCH + DCC Brownlees & Engle (2017); connectedness tables need bootstrap intervals and shrinkage VARs

Result from the data

- ΔCoVaR 1% of eleven banks: 4 of 25 within-region pairs differ at 5%, 0 after Holm; overlap of intervals alone would have ranked none
- Simulated LRMES (22 days, market below -10%): JPMorgan 10.6% today, 23.7% with the information of 31 March 2020

Leads to

- Stress testing and the team projects

Recap: Learning, Text and New Markets

- ▣ Flexible models need a strict process: purging, embargo, deflated performance measures
- ▣ Foundation models show no significant difference from strong volatility benchmarks and do not replace GARCH-based FHS for tail risk
- ▣ Text, crypto, bubbles and systemic risk reuse the same statistical toolkit
- ▣ New data and new models: the same five questions apply

Six Pitfalls That Returned in Every Chapter

Pitfalls of design

- ▣ **Look-ahead bias:** information from after t used at t
- ▣ **Overfitting:** a model tuned to the sample it is judged on
- ▣ **Multiple testing:** the best of many tries reported as if it were the only one

Pitfalls of data and inference

- ▣ **Wrong calendar join:** returns computed before aligning assets
- ▣ **Heavy tails ignored:** standard errors that assume the Normal distribution and independence
- ▣ **Wrong loss:** a loss function that cannot rank the forecast target

Pitfall 1: Look-Ahead Bias

- Any parameter, threshold or scaling estimated on the full sample leaks the future
 - ▶ Examples: full-sample GARCH parameters, full-sample standardisation, labels that overlap the test set
- BET, VaR 1%, 2005–2026: GARCH- t with full-sample parameters breaches on 1.31% of days, with rolling estimation on 1.47%
 - ▶ The look-ahead version looks better than any forecaster could have done in real time
- Chapter 13: shuffled K-fold reports 69.4% accuracy on pure noise
- Remedy: rolling or expanding windows; purging and embargo; write the information set $\mathcal{F}_{t-1}$ next to every forecast

Pitfall 2: Overfitting

- The in-sample fit rewards noise that will not repeat
 - ▶ Chapter 4: in-sample Sharpe 1.57, true Sharpe 0.30 for the same portfolios
- Backtest overfitting: trying many strategies and keeping the best [Bailey et al. \(2017\)](#)
 - ▶ The deflated Sharpe ratio corrects for the number of trials and for non-normal returns [Bailey & López de Prado \(2014\)](#)
- Remedy: fix the design before looking at the test period; report every model you tried; compare with a simple benchmark (1/N, historical simulation, random walk, log-HAR)

Pitfall 3: Multiple Testing

- Chapter 3: with 300 useless factors, on average 15.0 pass $|t| > 1.96$, and the best reaches $|t| \approx 3.06$
 - ▶ This is why the factor literature asks for $|t| > 3$ [Harvey, Liu & Zhu \(2016\)](#)
- Calendar anomalies, trading rules, model specifications: each extra test adds a chance of a false discovery
- Remedy: count your tests and correct
 - ▶ Bonferroni and Holm control the chance of any false discovery; Benjamini–Hochberg controls their share
 - ▶ For forecasts: the model confidence set instead of many pairwise tests [Hansen, Lunde & Nason \(2011\)](#)

Pitfall 4: Joining on Returns Instead of Prices

- ❑ Bitcoin trades every day, SPY only on weekdays
 - ▶ Wrong: compute each return on its own calendar, then keep the common days: Bitcoin's weekend moves disappear
 - ▶ Right: keep the common days of the **prices**, then compute returns: Monday covers Friday to Monday for both
- ❑ Bitcoin, 2020–2026, on the common days: cumulative return 1025% (right) against 680% (wrong)
 - ▶ Monday standard deviation 5.04% (right) against 3.83% (wrong): the wrong join hides weekend risk
- ❑ Remedy: align prices on common days first; one series alone stays on its own calendar

Pitfall 5: Heavy Tails Ignored in Inference

- Classical standard errors assume independence and finite fourth moments
 - ▶ BET losses: Hill tail index $\hat{\alpha} = 2.39$, block-bootstrap 95% CI [2.12, 3.03] [Hill \(1975\)](#): with $\alpha < 4$ the population kurtosis is infinite
 - ▶ So the sample kurtosis (11.1) estimates nothing, and a bootstrap interval for it is not valid [Athreya \(1987\)](#): report the tail index instead
- BET GARCH persistence: the Wald CI [0.974, 1.007] crosses the stationarity boundary $\alpha + \beta = 1$, where the Normal approximation fails [Andrews \(1999\)](#)
 - ▶ Profile-likelihood CI inside the parameter space $p \leq 1$: [0.975, 1] (IGARCH, $p = 1$, not rejected against the boundary critical value 2.71 efAndrewsB), so the half-life is at least 27 days and has no finite upper bound
- Remedy: robust (HAC) errors, block bootstrap where it is valid, tail indices instead of kurtosis, boundary-aware intervals

Pitfall 6: A Loss Function That Cannot Rank the Target

- Variance forecasts are compared with a noisy proxy (r_t^2 or RV_t)
 - ▶ With a conditionally unbiased proxy of the conditional variance (e.g. r_t^2 of a zero-mean return, or RV), MSE and QLIKE rank forecasts by expected loss as the true variance would; QLIKE depends only on the ratio proxy/forecast, so volatile days weigh less than under MSE [Patton \(2011\)](#)
 - ▶ Chapter 9: log-HAR beats GARCH in QLIKE; the MSE comparison is inconclusive
- ES alone has no strictly consistent scoring function on the usual broad classes of distributions; the pair (VaR, ES) has one, under mild conditions [Fissler & Ziegel \(2016\)](#)
 - ▶ Use FZ0 with Diebold–Mariano tests [Diebold & Mariano \(1995\)](#); expect finite-sample limits on precision [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- Remedy: choose the loss before the comparison, and state it in the project report

Recap: A Checklist Against the Pitfalls

Design

- ▣ Every forecast uses only $\mathcal{F}_{t-1}$
- ▣ Test period fixed in advance and used once
- ▣ All models tried are reported; tests are counted

Data and inference

- ▣ Prices joined before returns; adjustments and calendars documented
- ▣ Robust or bootstrap uncertainty for every headline number
- ▣ Loss function consistent for the target; comparison with a simple benchmark

One Question, Four Chapters

The question

- How much can a BET index position lose tomorrow, with probability 1%?
- Which forecast of that loss would we have trusted in real time, 2005–2026?

Data

- BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6 670 daily log returns
- Duplicate records (rows identical to the previous one) and weekend-dated rows removed; trading days with an unchanged close kept

The four steps

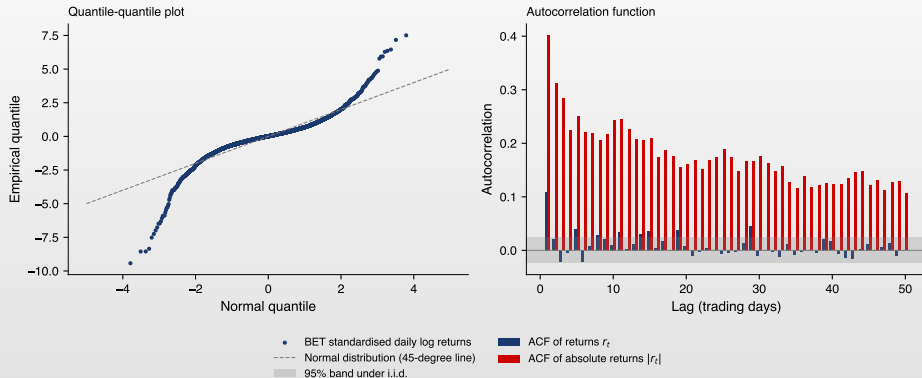
- Chapter 1: stylised facts, with robust inference
- Chapter 5: GARCH(1,1) with Student- t innovations
- Chapter 7: one-day VaR 1% from four models
- Chapter 8: coverage, independence and Basel zones



The Stock Exchange Palace, Bucharest (2023)

Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

Step 1: Stylised Facts of the BET

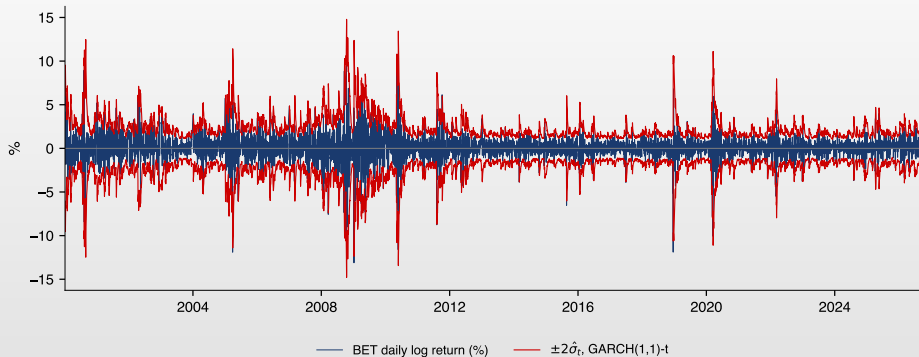


□ Left: quantiles of standardised returns against the Normal distribution; right: autocorrelations of r_t and $|r_t|$, lags 1–50

Reading Step 1

- Heavy tails: Hill index of losses $\hat{\alpha} = 2.39$ (block-bootstrap 95% CI [2.12, 3.03], $k = 153$); below 4, so kurtosis is not reported as an estimate
 - ▶ 51 days beyond ± 4 standard deviations, against 0.42 expected under the Normal distribution
 - ▶ Worst day: -13.1% (log) on 7 Jan 2009
- Clustering: first-order autocorrelation of $|r_t|$ is 0.40 and decays slowly; Ljung–Box $Q(10)$ on r_t^2 is 2440
 - ▶ Returns themselves have first-order autocorrelation 0.11: the BET is less liquid than large markets (Chapters 2, 10)
- Consequence: a Normal, constant-volatility VaR will be too low in stress and too high in calm periods

Step 2: GARCH(1,1)- t Volatility



□ BET daily log returns with $\pm 2\hat{\sigma}_t$ from a GARCH(1,1) with Student- t innovations, estimated on 2000–2026

Reading Step 2

- ▣ Estimates (robust standard errors): $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$
 - ▶ $\hat{\nu} \approx 5$: even after filtering, shocks have heavy tails
- ▣ Persistence $\hat{\alpha} + \hat{\beta} = 0.9907$: a half-life of about 74 trading days
 - ▶ Profile-likelihood 95% CI [0.975, 1]: persistence cannot be bounded away from 1 (Pitfall 5)
- ▣ Peak daily volatility 7.39% on 15 Oct 2008; at the end of the sample 1.67%

Step 3: Four VaR 1% Forecasts

Unconditional (last 500 days)

- ▣ **HS**: the loss $L = -r$ exceeded on 1% of the last 500 days (empirical quantile)
- ▣ **Normal**: $\text{VaR} = -(\hat{\mu} + \hat{\sigma} z_{0.01})$, $z_{0.01} = -2.326$

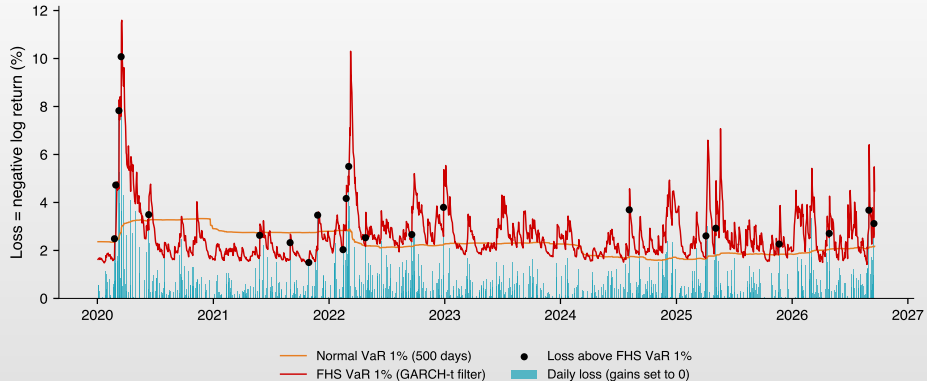
Conditional (GARCH- t on the last 1000 days)

- ▣ **GARCH- t** : $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t t_\nu^{-1}(0.01) \sqrt{(\nu - 2)/\nu})$
- ▣ **FHS**: $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \hat{q}_{0.01}(z))$, the 1% quantile of the standardised residuals [McNeil & Frey \(2000\)](#)

Real-time design

- ▣ Forecast for day t uses data up to $t - 1$ only
- ▣ GARCH re-estimated every 21 trading days; $\hat{\sigma}_t$ updated daily
- ▣ Evaluation: 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days
- ▣ Expected breaches at 1%: about 54

Step 3: Losses and VaR 1%, 2020–2026

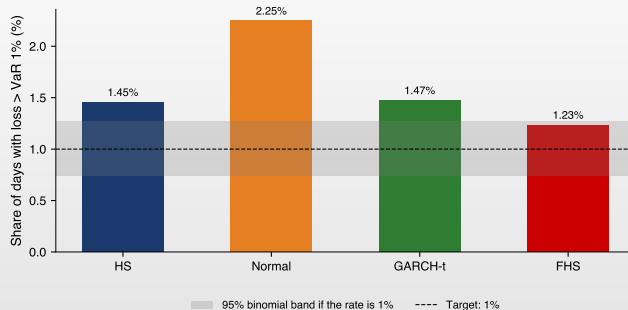


□ Daily BET losses (gains set to 0), Normal and FHS VaR 1%; black dots: losses above the FHS VaR

Reading Step 3

- FHS reacts within days: it peaked at 11.6% on 18 Mar 2020, during the COVID-19 crash
 - ▶ The Normal VaR on a 500-day window moves only slowly
- 2020–2026 (1 675 days, about 17 breaches expected): FHS 22 breaches, Normal 41
 - ▶ Normal breaches cluster in the first weeks of every stress episode
- Last forecast in the sample: FHS 4.48%, Normal 2.18%: the two models disagree most when volatility changes

Step 4: Breach Rates



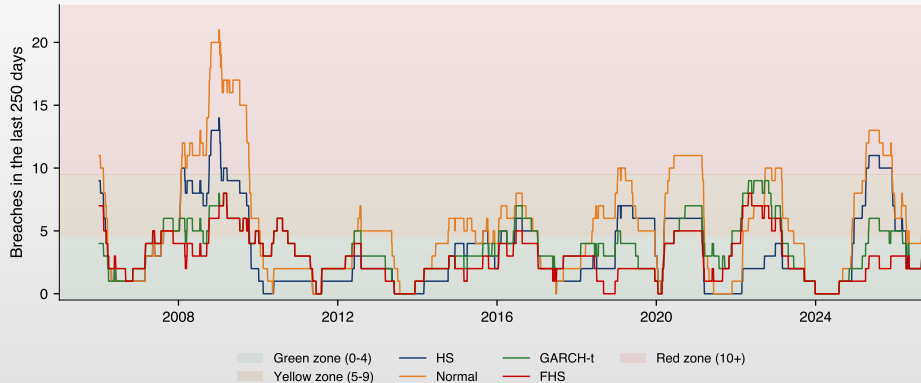
- Share of days with a loss above the VaR 1% forecast, 2005–2026; grey band: 95% binomial range if the true rate is 1%

Reading Step 4: Coverage and Independence

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

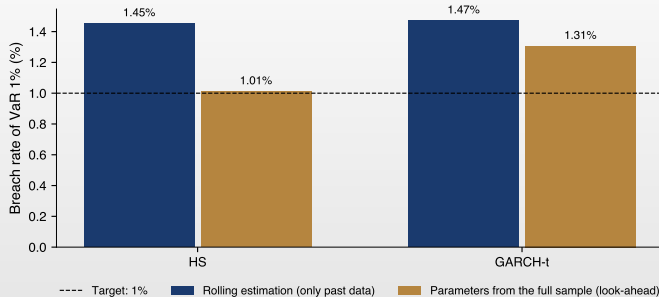
- ▣ p_{uc} : Kupiec coverage test; p_{ind} : Christoffersen independence test; p_{cc} : both jointly [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#)
- ▣ Red zone: share of 250-day windows with 10 or more breaches (Basel traffic light)
- ▣ Only FHS is rejected by none of the three tests at 5%; HS and Normal also fail independence: their breaches cluster
- ▣ GARCH-t: independence is not rejected, but there are too many breaches: the Student-t tail is not heavy enough at 1%

Step 4: Breaches in the Last 250 Days



- Number of breaches in a moving 250-day window, with the Basel green, yellow and red zones
- The Normal model spends 18.6% of the windows in the red zone, HS 5.5%, the two GARCH-filtered models never

The Look-Ahead Trap in the Case Study



- Full-sample HS quantile: 1.01% breaches, by construction; full-sample GARCH-t: 1.31%
- Both look better than their real-time versions: the gap is the value of information nobody had

Step 5: Comparing the Four Models Formally

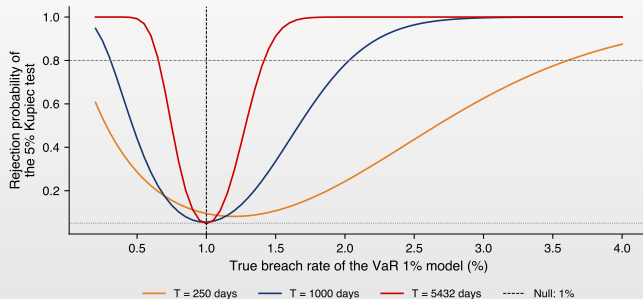
- **Dynamic Quantile** test: regress $I_t - 0.01$ on a constant, four lagged hits and VaR_t ; $DQ \sim \chi_6^2$ Engle & Manganelli (2004)
 - ▶ It detects breaches that are predictable from the past or from the level of VaR, which the Christoffersen test (one lag) can miss
- **FZ0 score** for the pair (VaR 2.5%, ES 2.5%), $v = -\text{VaR}$, $e = -\text{ES}$, return y : $S = -\frac{\mathbf{1}\{y \leq v\}(v-y)}{\alpha e} + \frac{v}{e} + \ln(-e) - 1$ Patton, Ziegel & Chen (2019)
 - ▶ Consistent for (VaR, ES) Fissler & Ziegel (2016); pairwise Diebold–Mariano tests with HAC variance Diebold & Mariano (1995); the model confidence set at 10% Hansen, Lunde & Nason (2011)
- **ES backtest**: comparative, against HS as the standard model; red if the model is significantly worse, green if significantly better, yellow otherwise Nolde & Ziegel (2017)
 - ▶ All forecasts use only data up to $t - 1$; 5 432 days, 3 Jan 2005 – 18 Sep 2026

Step 5: Results

Model	DQ	p_{DQ}	FZ0	p_{MCS}	ES zone
HS	192.1	<0.001	1.424	0.00	standard
Normal	364.6	<0.001	1.495	0.00	red
GARCH-t	17.9	0.007	1.175	0.18	green
FHS	9.9	0.129	1.157	1.00	green

- Only FHS passes the DQ test at 5% ($p = 0.129$); GARCH- t fails it ($p = 0.007$) through its excess of breaches
- FZ0: FHS and GARCH- t are not significantly different (DM $t = 1.39$), both beat HS and Normal ($|t| > 4.9$); the 10% model confidence set is {GARCH- t , FHS}
- Comparative ES backtest: Normal red, GARCH- t and FHS green: the ranking of Step 4 survives formal inference

How Much Can a Backtest Detect? The Power of Kupiec



- Exact binomial probability that the 5% Kupiec test rejects a VaR 1% model, as a function of its true breach rate, for $T = 250, 1000$ and 5432 days

Reading the Power Curves

- $T = 250$ (one Basel year): a true rate of 2% is detected only 24% of the time; rejection needs 0 or at least 7 breaches
 - ▶ Minimum detectable rate with 80% power: 3.65% at $T = 250$, 2.05% at $T = 1\,000$, 1.45% at $T = 5\,432$
- Our case study, $T = 5\,432$: a true rate of 1.3% would be detected with probability 55% only
- A non-rejection with low power is “no evidence”, not “no problem”: state the power before the test

Recap: The Case Study

- ▣ The BET shows every stylised fact: heavy tails, clustering, some linear autocorrelation
- ▣ GARCH- t captures clustering; the filtered residuals still have heavy tails
- ▣ A GARCH filter with an empirical tail (FHS) is the only VaR 1% that passes coverage, independence and the DQ test; with GARCH- t it forms the model confidence set for (VaR, ES)
- ▣ Real-time design matters: full-sample estimates flatter every model
- ▣ The same four steps are a template for your project

What the Project Is

- An empirical study of one question about financial markets, answered with the methods of this course
 - ▶ Teams; four steps: replicate, extend, deliver on GitHub, present and defend (next slide)
- It carries **70%** of the final grade; quizzes and seminar activity carry 20%, attendance 10%
- Starting points: the Part C question of every seminar (next section)
 - ▶ Your own question is welcome if it passes the five questions of this chapter
- A good project can end with a negative answer, if the evidence is honest and precise

The Project in Four Steps

- 1. **Replicate**: reproduce one full table or figure of a published paper (or of the lecture), cell by cell
 - ▶ Report the discrepancy of every cell and explain it; an AI assistant cannot guess a published table
- 2. **Extend**: new data, a new market (BVB, crypto) or a more recent sample, under a **pre-analysis plan** committed before the test period is opened
 - ▶ The plan states the null, the test, the loss, the sample split, the identification strategy, the power of the main test and the specification grid with its correction
- 3. **Deliver on GitHub**: code, lockfile, fixed seeds, data provenance; one notebook regenerates every number
 - ▶ `AI_USE.md` and `AI_ERRORS.md` are mandatory; the report is an 8–12-page paper
- 4. **Present and defend**: team presentation, then a 10-minute oral defence
 - ▶ Each member explains one result and answers a “what changes if...” question; grades can differ between members

Replication Standard

- Target: one full table or figure, with its DOI, page and table number, not a single coefficient
 - ▶ Report a cell-by-cell comparison: published value, your value, difference, and the reason for each gap (sample, filter, data vendor, definition)
- Large-scale replications show why: many anomalies shrink out of sample or with better data [Hou, Xue & Zhang \(2020\)](#), while most factors replicate once the design is right [Jensen, Kelly & Pedersen \(2023\)](#)
- Reproducibility: a lockfile of package versions, fixed random seeds, series names and dates of the data, one command that regenerates every number
 - ▶ Different teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#): reproducible code narrows the gap

Identification: from Correlation to a Claim

- Event questions (“did spot ETFs, the FTSE upgrade or a bank tax change X?”) need a counterfactual
 - ▶ A control market not hit by the event: difference-in-differences on the risk measure
 - ▶ A synthetic control built from several untreated markets [Abadie, Diamond & Hainmueller \(2010\)](#); placebo event dates (e.g. $\pm$ one year)
- Unknown timing: a sup-Wald break test [Andrews \(1993\)](#); report the estimated break date, not an imposed one
- Forecasting questions: out of sample only, with a test period declared in the plan
- One event is $n = 1$: say what the design can and cannot identify

Power and Minimum Detectable Effect

- Before running the main test, compute the smallest effect the design can detect with 80% power at 5%
 - ▶ Kupiec on 250 days cannot distinguish a VaR 1% model from a 2% one: power 24%; minimum detectable rate 3.65% (Step 5)
- Deliverable: a simulated power curve of the main test at your T , under the null and under the alternative you care about
- Short samples of crypto, ETFs or post-event windows make this binding: many project questions are underpowered
- A non-rejection with low power is “no evidence”, not “no effect”

Robustness without p-Hacking

- The pre-analysis plan is a timestamped commit in the repository, made before the test period is opened
- Specification curve: estimate every defensible combination of window, lags, α and sample start; plot all estimates sorted, with their intervals [Simonsohn, Simmons & Nelson \(2020\)](#)
 - ▶ Report the spread across specifications, not the best cell
- Correct the family: Holm for any false discovery, Benjamini–Hochberg for their share, or a bootstrap reality check
 - ▶ False and missed discoveries in finance [Harvey & Liu \(2020\)](#); the scientific outlook of the field [Harvey \(2017\)](#)

Forecast Evaluation for Risk Projects

- Volatility: compare forecasts with a noisy, conditionally unbiased proxy only through a robust loss (MSE or QLIKE) [Patton \(2011\)](#)
 - ▶ Many models: the model confidence set [Hansen, Lunde & Nason \(2011\)](#), not the lowest average loss
- VaR: coverage, independence and the DQ test [Engle & Manganelli \(2004\)](#); with a benchmark, conditional predictive ability tests [Giacomini & White \(2006\)](#)
 - ▶ State the power of every backtest at your T
- ES: score the pair (VaR, ES) with FZ0; comparative backtests with three zones [Nolde & Ziegel \(2017\)](#); dynamic (VaR, ES) models [Patton, Ziegel & Chen \(2019\)](#)
 - ▶ The integrated case study (Steps 1–5) is the template

Choosing the Question

A good question is

- **Specific:** one market, one period, one quantity
- **Testable:** a null hypothesis and a benchmark
- **Feasible:** data you can obtain and clean in a few weeks
- **Relevant:** an answer matters to an investor, a bank or a regulator

Examples

- Too vague: "Is Bitcoin risky?"
 - ▶ Better: "Did spot ETFs change the one-day VaR 1% of Bitcoin, and which model forecasts it best since 2024?"
- Too broad: "Can machine learning predict the market?"
 - ▶ Better: "Does a boosting model beat log-HAR for next-day BET volatility, in QLIKE, 2018–2026?"

Data

- Use the course data where possible: daily series of indices, ETFs, BVB stocks, crypto and FX, plus intraday bars
 - ▶ Public sources: BNR reference rates, FRED, Kenneth French Data Library, DefiLlama; the BVB archive for Romanian indices
- Apply the course conventions
 - ▶ Adjusted prices for ETFs and stocks, closing values for indices, FX and crypto
 - ▶ Remove holiday fills and weekend quotes where the market is closed; annualise with the real frequency
 - ▶ Several assets: join on prices, then compute returns
- Look for bad prints before modelling: jumps of $\pm 50\%$ in adjusted prices are usually data errors

Methods: From Question to Chapter

If your question is about	Start from
Return distribution, tails, predictability Expected returns, factors, portfolios	Chapters 1–2: moments, robust tests, variance ratios Chapters 3–4: Fama–MacBeth, shrinkage, out-of-sample Sharpe tests
Volatility and correlation Risk measures and capital	Chapters 5, 6, 9: GARCH, DCC, copulas, log-HAR Chapters 7–8: FHS, EVT, backtests, FZ0
Liquidity, options, continuous time	Chapters 10–12: Amihud, Black–Scholes, VIX, variance risk premium
Forecasting with ML, deep learning or text Crypto, bubbles, systemic risk	Chapters 13–15: purged CV, foundation models, sentiment Chapters 16–18: stablecoins, GSADF, CoVaR, networks

Honest Evaluation Checklist

Before you look at the results

- ☐ Write the hypothesis, the benchmark and the loss function
- ☐ Fix the estimation window and the test period
- ☐ Decide how many models and parameters you will try, and the correction for them
- ☐ Write the identification strategy and the power of the main test

When you report

- ☐ Out-of-sample results only for claims about forecasting
- ☐ Every headline number with a standard error, CI or p -value
- ☐ Costs for trading strategies; subperiods and a second market for robustness
- ☐ What did **not** work, and why

The Report: an 8–12-Page Paper

- **1. Introduction:** question, hypothesis and contribution; **2. Literature:** 10–15 clickable references, at least 3 from the last five years, including [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#)
- **3. Data and replication:** sources, period, cleaning rules; the replicated table cell by cell
- **4. Design:** the pre-analysis plan, identification strategy, power, benchmark, loss and tests
- **5. Results:** tables and charts with uncertainty; **6. Robustness:** the specification curve with the multiplicity correction; **7. Conclusion:** the answer in one sentence, and its limits
- **Code:** a GitHub repository whose README and notebook reproduce every number, with [AI_USE.md](#) and [AI_ERRORS.md](#)

The Presentation

Structure (about 12 minutes)

- ▣ The question and why it matters (1 slide)
- ▣ Data and one chart that shows the problem (2 slides)
- ▣ Method and benchmark (2 slides)
- ▣ Main result with its uncertainty (2–3 slides)
- ▣ Robustness, limitations, answer (2 slides)

Good practice

- ▣ One message per slide; charts with readable axes and the legend below
- ▣ Every team member presents and can answer questions on the code
- ▣ Then a 10-minute oral defence: each member explains one result and answers a “what changes if...” question, without AI

Project Grading (70% of the Final Grade)

Component	What we look for
Replication (step 1)	A published table or figure reproduced cell by cell; every gap explained
Question, data and extension (step 2)	Specific, testable question; documented data and calendar alignment
Methods and inference	Pre-analysis plan, identification strategy, power, real-time design, benchmark, specification curve with a multiplicity correction
Interpretation and report	Economic meaning, limitations, honest conclusion; an 8–12-page paper with 10–15 references
GitHub repository (step 3)	One command reproduces every result (lockfile, seeds); AI_USE.md, AI_ERRORS.md
Presentation and oral defence (step 4)	Individual: one result explained, answers to “what changes if...”
Total	70% of the final grade; plus quizzes and activity (20%) and attendance (10%)

- Grades can differ between team members, depending on the oral defence

Timeline

- Week 1 after this chapter: team and question
 - ▶ Submit a one-page proposal (template in Seminar 19, Part C); open the GitHub repository
- Weeks 2–3: data, the replication of one published table or figure (step 1) and the pre-analysis plan committed to the repository
- Weeks 4–5: extension, evaluation, robustness (step 2); feedback at the seminar
- Final weeks: repository with README and the two AI files (step 3); presentation and oral defence (step 4)
- Exact dates are announced at the course and seminar

Where Projects Usually Lose Points

- ▣ A question that changes after the results are known
- ▣ In-sample fit presented as forecasting power
- ▣ No benchmark, or a weak one chosen to be beaten
- ▣ Numbers without uncertainty; many tests without correction
- ▣ Charts without units, legends inside the data, unexplained acronyms
- ▣ Code that does not run, or numbers in the report that the code does not produce
- ▣ AI output kept without checks: undeclared use, unchecked code, references that do not exist

Recap: The Team Project

- ▣ One specific, testable question; one benchmark; one loss function chosen in advance
- ▣ Clean data with the course conventions; real-time design; honest reporting
- ▣ Replicate, extend, deliver on GitHub, present and defend: 70% of the grade

The AI Policy of the Course

- ▣ **Allowed:** AI assistants for code, debugging, literature search and writing, in the project and in the seminars
- ▣ **Declared:** every tool, what it was used for and what you checked yourselves, in [AI_USE.md](#)
 - ▶ Undeclared use of AI counts as plagiarism
- ▣ **Checked:** you are responsible for every line of code and text
 - ▶ “The AI wrote it” is not an explanation at the oral defence
 - ▶ Every reference needs a working DOI or link; a reference that does not exist counts as fabricated data
- ▣ **No AI during the oral defence:** you answer alone, from your own understanding of the project

The Two Mandatory Files

AI_USE.md: the declaration

- ▣ Tools: name, version, who used them and for what
- ▣ Files or sections where AI-written code or text was kept, and what you changed
- ▣ How you verified the replicated table, the data, the formulas and the references
- ▣ What the team did without AI

▣ Templates in English and Romanian: github.com/danpele/MFM/tree/main/project

AI_ERRORS.md: the error log

- ▣ At least **three** places where the assistant was wrong
- ▣ For each: what you asked, what it answered, what was wrong, how you found out, the correct version
- ▣ Errors you caught are a strength of the project, not a weakness

Typical AI Errors in Financial Modelling (1/2)

- ▣ **Look-ahead bias:** a rolling statistic or a scaler that includes day t in the forecast for day t
 - ▶ Check: change the last observation; the forecast for that day must not move; shift by one day (`.shift(1)`)
- ▣ **Sign of VaR:** the 1% quantile of returns reported as VaR, or the 99% quantile of returns (a gain) used as the loss
 - ▶ Check: $\text{VaR}_\alpha = -q_\alpha(r) > 0$ for a loss; one-day VaR 1% for an index is a few percent, not negative
- ▣ **Annualisation:** daily volatility multiplied by 252 instead of $\sqrt{252}$, or 252 days used for crypto that trades every day
 - ▶ Check: means scale with h , volatilities with $\sqrt{h}$; use the real number of trading days (365 for crypto)

Typical AI Errors in Financial Modelling (2/2)

- **Random cross-validation (CV) on time series:** shuffled K-fold trains on the future; in Chapter 13 it reports 69.4% accuracy on pure noise
 - ▶ Check: is any test date earlier than a training date? Use walk-forward or purged CV with an embargo [López de Prado \(2018\)](#)
- **Invented references:** plausible authors, title and journal, but the paper does not exist
 - ▶ Check: search every DOI on [Crossref](#); open the paper and find the claim you cite
- **Also frequent:** wrong degrees of freedom in a test, library defaults you did not choose, numbers stated without running any code
 - ▶ Every seminar has a C2 exercise that trains you to catch such errors

Recap: Using AI in Your Project

- Use AI freely, declare it in `AI_USE.md`, and check every line it writes
- The replicated table is your best test: an AI cannot guess it
- Log at least three AI errors in `AI_ERRORS.md`; at the oral defence you answer alone

Project Seeds: Part C of Every Seminar (1/2)

Ch.	Open question (link to the seminar)
0	Did spot ETFs change Bitcoin's correlation with equities?
1	Did BET-TR autocorrelation fall after the FTSE reclassification?
2	Did the BVB become more efficient after the 2020 FTSE upgrade?
3	Do factor premia exist on the Bucharest Stock Exchange?
4	An optimal portfolio of BVB blue chips in RON
5	Is the Romanian market's volatility driven by the US?
6	Did Romanian and euro-area equities become more integrated after 2020?
7	Capital for a BVB portfolio vs an S&P 500 portfolio

- Each seminar gives the data, a starting method and a reference analysis discussed in class

Project Seeds: Part C of Every Seminar (2/2)

Ch.	Open question (link to the seminar)
8	Which risk model would you certify for a Romanian bank's trading book?
9	Is Bitcoin's volatility more predictable than the S&P 500's?
10	Does illiquidity carry a premium on the BVB?
11	Which continuous-time model fits BET and Bitcoin?
12	Is the variance risk premium a tradable signal?
13	Does a foundation model beat boosting?
14	Can a zero-shot foundation model replace GARCH for the BET?
15	Does headline sentiment pay after costs?
16	Is Bitcoin an alternative asset, a risk asset or a hedge?
17	Is there an AI bubble?
18	How exposed are Romanian banks to euro-area banking stress?

- Chapter 20 (realised volatility with signatures, VOLARE data) also suits a forecasting project

Turning a Seed into a Project

- Replicate first: one published table or figure from a paper behind the seed (step 1)
- Narrow it: one market, one period, one number to forecast or test
 - ▶ Example: Seminar 8 becomes “Which of HS, FHS and GARCH-EVT gives a correct VaR 1% for a 50/50 BET–EUR portfolio in 2015–2026?”
- Extend it: a second market, a new model, a longer or more recent sample
 - ▶ Example: add a foundation model (Chapter 14) as a volatility filter inside FHS
- Keep the reference analysis as your benchmark and try to improve on it honestly

Research Question: How Much of a Result Is the Researcher?

- Menkveld et al. (2024), *Journal of Finance* 79(3): 164 research teams test the same six hypotheses on the same data (Finance Crowd Analysis Project, #fincap; Sec. I.A)
 - ▶ **Standard error**: uncertainty from sampling, the data-generating process; **nonstandard error**: uncertainty from the analysis path each researcher picks, the evidence-generating process (Introduction)
- **Model**: team j reports $X_j = e_j + \varepsilon_j$, with e_j the researcher fixed effect of its path and ε_j the sampling error (Sec. I.C, eq. 1)
 - ▶ Nonstandard error $:= Q_{0.75}(x) - Q_{0.25}(x)$, the interquartile range (IQR) of the estimates across teams (eq. 3); the IQR, not the standard deviation, because the e_j can have fat tails
- **Four questions**: how large is it; is it smaller for better and more reproducible work; does peer feedback reduce it; do researchers know its size? (Introduction)
 - ▶ Link to this chapter: every team project is one path through the “garden of forking paths”; pre-registration, the specification grid and a repository that reruns every number are the remedies

Data and Method

- **Data:** all trades in the Euro Stoxx 50 index futures, 2002–2018, 720 million records, each side flagged as client or own account (Sec. I.A)
 - ▶ Six hypotheses, each “no annual trend” in: market efficiency, realised bid–ask spread, client share of volume, realised spread on client orders, share of market orders among client orders, gross trading revenue of clients
 - ▶ Each team chooses its own measure and reports the average yearly change in % with its standard error (SE)
- **Four stages** in 2021: independent analysis (paper and code); feedback from two peer evaluators; the five best papers shared; final estimates without the code constraint (Sec. I.A)
 - ▶ A pre-analysis plan was filed publicly before the experiment started (Sec. I.A)
- **Quality:** team quality (first principal component of five proxies), reproducibility score 0–100 by the certification agency Cascad, demeaned peer rating (Sec. I.B)
 - ▶ Quantile regressions of the estimates at the 10th, 25th, 50th, 75th and 90th percentiles; a shrinking IQR means less dispersion (Sec. I.C)
- **Inference:** each estimate is tested against the median estimate, with a Bonferroni correction at a family level of 0.5% (Sec. I.C)
 - ▶ Multiverse analysis: the teams report their choice at each fork of Table V; the k -sample Anderson–Darling statistic ranks the forks by how much they move the estimates (Sec. II.C)

Key Results

- **Size:** for market efficiency the median estimate is -1.1% per year, with an IQR of 6.7 percentage points (pp); even for the client share of volume, a simple ratio, the IQR is 1.2 pp around -3.3% (Table I, Panel C)
 - ▶ Interpretation: the median SE of 2.5% implies a Gaussian IQR of $1.35 \times 2.5 = 3.4\%$, half the 6.7 pp; the no-dispersion null is rejected for all six hypotheses, $p < 0.0001$ (Table II)
- **Quality:** a rise of one standard deviation in reproducibility lowers the IQR by 25.0%, in peer rating by 33.3% (Table III, Fig. 2)
 - ▶ Interpretation: code that reruns end to end is associated with answers closer to the consensus
- **Feedback:** from stage 1 to stage 4 the IQR falls by 47.2% and the interdecile range by 68.2% (Table IV, Fig. 3)
 - ▶ Interpretation: peer review narrows the dispersion but does not remove it
- **Beliefs:** pooled over hypotheses, the teams' predicted standard deviation of estimates is 71.7% below the realised one (Sec. II.B.3, Fig. 4)
 - ▶ Interpretation: researchers are overconfident about how unique their own path is
- **Forks:** model choice and sampling frequency move the estimates most; a relative-change model gives a median annual trend of $-23\,000\%$ on daily data, -200% monthly and -4.56% yearly (Fig. 6, Fig. 7)
 - ▶ Interpretation: Jensen's inequality, $\prod_{t=1}^T \mathbb{E}(M_t) < \mathbb{E}(\prod_{t=1}^T M_t)$ (eq. 7); log differences and linear trends do not suffer from it

Replicate on Course Data, Limits, Extension

- **Replication:** a multiverse for the market-efficiency hypothesis on the Euro Stoxx 50 index and the BET (data/market), 2002–2018, then extended to the last full year (Seminar 19, C3)
 - ▶ Forks from Table V: outliers (none, winsorised or trimmed at the 2.5 and 97.5 percentiles), measure (variance ratio, R^2 of an AR model), frequencies (day to week, day to month), model (linear trend, log difference, relative difference)
 - ▶ Data limitation: daily closes only, so the intraday frequency pairs of Table V cannot be run
- **Limits stated by the authors:** the multiverse covers a few forks only and explains the reported estimates poorly (footnote 32); beliefs were asked as standard deviations, not IQRs (footnote 28)
 - ▶ The test of dispersion assumes consistent reported SEs (Sec. I.C); the dynamics of journal refereeing are kept out of scope (Sec. I.A)
- **Open extension:** does a pre-registered grid shrink the nonstandard error of a risk question? Measure the IQR of VaR 1% conclusions for the BET across defensible paths
 - ▶ For your project: report the distribution over your grid and the forks that move it most, not a single cell

An Open Question

How much of a result is the data, and how much the researcher?

- 164 teams tested the same hypotheses on the same data: the dispersion of their answers, the **nonstandard errors**, was sizable and underestimated by the teams [Menkveld et al. \(2024\)](#)
 - ▶ It was smaller for more reproducible research and fell after peer-review stages
- Replication is not hopeless: in a Bayesian model of replication most asset-pricing factors replicate and work out of sample in 93 countries [Jensen, Kelly & Pedersen \(2023\)](#)
- AI assistants make every extra specification almost free: the forking paths multiply, but so does the ability to report all of them
 - ▶ Formal version for a project: the conclusion must hold over a pre-registered set S of defensible specifications, with the family corrected (FWER or FDR); report its distribution over S , not the best cell

The Discovery Loop, Step by Step (1/2)

- **Week 1, the question:** sharpen a seed into H_0/H_1 with a replication target
 - ▶ “Here is our seed from Part C of Seminar N. Give three sharper, testable versions, the published table each would replicate and the series in data/market it needs.”
- **Weeks 1–2, literature and target:** the number to replicate is one you have seen in the PDF
 - ▶ “List the most cited and the most recent papers on this question with DOI; for each, the headline estimate, its table and its sample.”
 - ▶ Resolve every DOI on [Crossref](#); search with Semantic Scholar, Elicit or Google Scholar; open the table yourselves
- **Week 2, hypotheses and pre-registration,** committed before the first result
 - ▶ “For our H_0 and H_1 , list competing hypotheses and the implication that separates each; then draft a pre-registration: sample, filters, windows, tests, correction.”
 - ▶ The commit history of the repository is the evidence that the plan came first

The Discovery Loop, Step by Step (2/2)

- **Weeks 2–3, code and replication:** the replicated table becomes a test in the repository
 - ▶ “Write a function that reproduces every cell of the published table from data/market and an assertion with the tolerance we accept.”
- **Weeks 4–5, extension and robustness:** run the whole pre-registered grid
 - ▶ “Here are all the cells we ran. Which conclusions survive a Holm correction, and which ones would a referee call data snooping?”
- **Before the defence, critique:** the assistant asks, the team answers
 - ▶ “Act as the examiner of our oral defence: ask the ten hardest questions on identification, inference and data. Do not answer them.”
 - ▶ The team answers without AI, as at the defence itself
- **Report:** the distribution of results over the grid, not the best cell; prompts in AI_USE.md, caught errors in AI_ERRORS.md

What the Researcher Must Check

- **The replication target exists:** number, table and sample taken from the paper, not from the assistant's summary
 - ▶ Test: the report cites the page and the table; a second member finds the same number there
- **The plan came first**
 - ▶ Test: in the commit history, the pre-registration precedes the first results file
- **Forking paths:** more specifications tried than reported
 - ▶ Test: every run writes its configuration and result to a log; the report states their number and corrects the family
- **Reproducibility:** numbers entered manually in the report
 - ▶ Test: a fresh clone and one command regenerate every number to the reported digit
- **Understanding:** each member can derive the key formula and defend every choice
 - ▶ Test: an internal mock defence, without AI, one week before the real one

Mini-Case: Replicate, Then Extend

- **Replicate:** FHS VaR 1% for the BET; Chapter 8 reports 59 breaches in 4 937 days (1.20%, Kupiec $p = 0.181$)
 - ▶ The case study of this chapter: 67 in 5 432 days (1.23%, $p = 0.095$); same model, different evaluation start (2007 vs 2005) and settings: document every gap before calling a replication failed
- A typical assistant proposal, reading each p -value alone at 5%: “only GARCH-EVT is correctly calibrated on all four assets”
- **Extend:** the 24 conditional-coverage tests of Chapter 8 (4 assets $\times$ 6 models) form one family
 - ▶ Rejections at 5%: 12 without correction, 10 with Holm (FWER), 11 with BH (FDR 5%)
 - ▶ FHS fails only on the S&P 500 ($p = 0.040$); within the family of 24 it survives, so after Holm FHS and GARCH-EVT pass on all four assets
- Interpretation: “FHS fails for the S&P 500” depends on whether the test is read alone or in its family; pre-register the family and report both

Your Turn: the Capstone Plan

- **Question:** your seed from Part C, written as H_0/H_1 , with the observation that would falsify it
- **Replicate first:** one published table seen in the paper, turned into a passing test in the repository
 - ▶ Then extend over a pre-registered grid, with the family corrected and the full distribution reported
- **Deliverables** along the timeline: proposal and replication target, pre-registration commit, passing replication test, grid of results, mock defence, report, oral defence
- **AI log:** the prompts that changed the design in AI_USE.md; every caught error, with the test that caught it, in AI_ERRORS.md ([templates](#))

Key Takeaways

- ▣ Returns are hard to predict; their risk is not: volatility, tails and dependence can be forecast
- ▣ Every model is judged out of sample, against a simple benchmark, with a loss that fits the target
- ▣ Heavy tails change inference: use robust errors and the bootstrap, report intervals
- ▣ Most wrong results come from the design, not from the model: look-ahead, overfitting, multiple testing, calendars
- ▣ New tools (ML, foundation models, LLMs) raise the bar for evaluation, not lower it

Further Study

Textbooks

- Franke, Härdle & Hafner (2019); Härdle, Chen & Overbeck (2017); McNeil, Frey & Embrechts (2015); Tsay (2010)

Advanced courses on quantinar.com

-  Statistics of Financial Markets
-  Measuring Statistical Risk

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_19

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